CO1 - AT2

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DSA0603 – Data Handling and Visualization

Slot A

QUESTION 1

Question 1: Correlation Heatmap [Python]

Dataset

Student	Attendance (%)	Study Hours/Day	Sleep Hours/Day	Marks (%)
S1	92	3.5	6.5	88
S2	75	1.5	8	62
S3	88	2.8	7	80
S4	60	1	5.5	48
S5	95	4	7.5	91
S6	70	2	6	58

Questions

1. Write a Python program (using pandas.corr() and seaborn.heatmap()) to compute and display the correlation matrix of Attendance, Study Hours, Sleep Hours, and Marks.

2. Identify the pair of variables with the strongest correlation.

3. Explain what a diverging color scale communicates in a correlation heatmap.

4. Discuss one limitation of using correlation to interpret this data.

5. Suggest one additional variable that could improve this analysis.

ANSWER 1

1.1 Python program

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

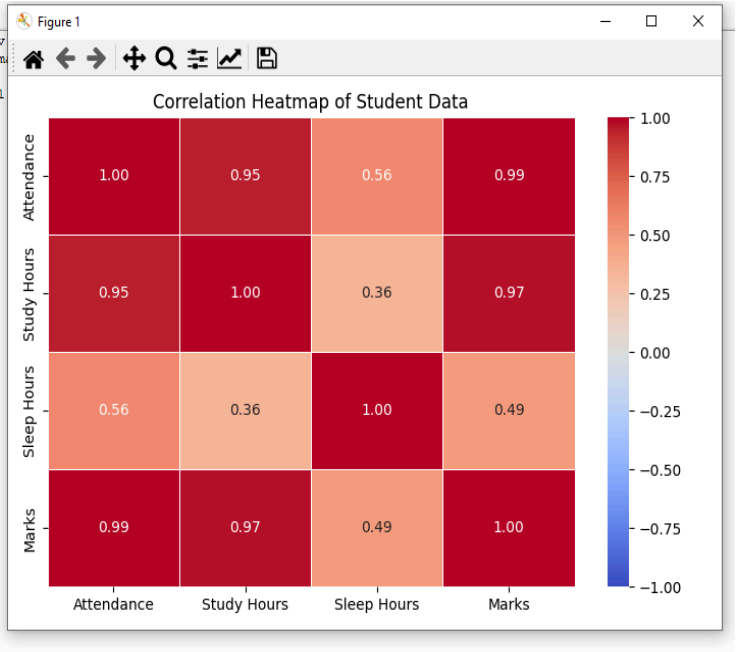
# Create the dataset
data = {
    "Student": ["S1", "S2", "S3", "S4", "S5", "S6"],
    "Attendance": [92, 75, 88, 60, 95, 70],
    "Study Hours": [3.5, 1.5, 2.8, 1.0, 4.0, 2.0],
    "Sleep Hours": [6.5, 8.0, 7.0, 5.5, 7.5, 6.0],
    "Marks": [88, 62, 80, 48, 91, 58]
}
df = pd.DataFrame(data)

# Compute the correlation matrix (numeric columns only)
corr = df[["Attendance", "Study Hours", "Sleep Hours", "Marks"]].corr()
print(corr.round(2))

# Display the heatmap
plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm",
            vmin=-1, vmax=1, linewidths=0.5)
plt.title("Correlation Heatmap of Student Data")
plt.tight_layout()
plt.show()
```

Output:

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun  6 2024, 19:30:16) [MSC v
Type "help", "copyright", "credits" or "license()" for more inform
>>>
========== RESTART: C:/Users/admin/OneDrive/Documents/DHDV/col - 1
Correlation Matrix:
             Attendance  Study Hours  Sleep Hours  Marks
Attendance         1.00         0.95         0.56   0.99
Study Hours        0.95         1.00         0.36   0.97
Sleep Hours        0.56         0.36         1.00   0.49
Marks              0.99         0.97         0.49   1.00
```



	Attendance	Study Hours	Sleep Hours	Marks
Attendance	1.00	0.95	0.56	0.99
Study Hours	0.95	1.00	0.36	0.97
Sleep Hours	0.56	0.36	1.00	0.49
Marks	0.99	0.97	0.49	1.00

1.2 Pair with the strongest correlation

Attendance and Marks have the strongest correlation ($r \approx 0.99$). Students with higher attendance almost always score higher marks. The next strongest pair is Study Hours and Marks ($r \approx 0.97$).

1.3 Diverging color scale

A diverging color scale uses two contrasting colors that meet at a neutral midpoint. In a correlation heatmap the midpoint is 0 (no correlation). One hue (for example red) shows positive correlation and the other (for example blue) shows negative correlation. The darker or more saturated the color, the stronger the relationship, and the pale middle shows values near zero. This lets the reader see both the direction and the strength of every correlation at a glance. Setting vmin=-1 and vmax=1 keeps the midpoint at exactly 0.

1.4 Limitation of correlation

Correlation does not prove causation. A high correlation between Attendance and Marks does not show that attendance causes higher marks; a third factor such as student motivation may influence both. Also, with only six students, the coefficients are unstable, and one student can change them greatly. Correlation measures only linear relationships and may miss other patterns (for example, sleep may help only up to an optimal point).

1.5 Additional variable

Previous academic performance (for example the previous semester's GPA or marks) would improve the analysis. It would help separate the effect of current habits (attendance, study hours) from the effect of a student's existing ability. Other useful choices are the number of assignments completed or the level of motivation.

QUESTION 2

Question 2: Nonlinear Axes [R]

Dataset

Week	App Downloads
1	50
2	95
3	180
4	400
5	820
6	1600
7	3100

Questions

1. Write an R program to display the data using a linear y-axis.
2. Modify the program to display the same data using a logarithmic y-axis (scale_y_log10()).
3. Compare both visualizations.
4. Explain why a logarithmic scale is suitable for this dataset.
5. Identify one situation where a logarithmic scale should not be used.

ANSWER 2

2.1 R program with a linear y-axis

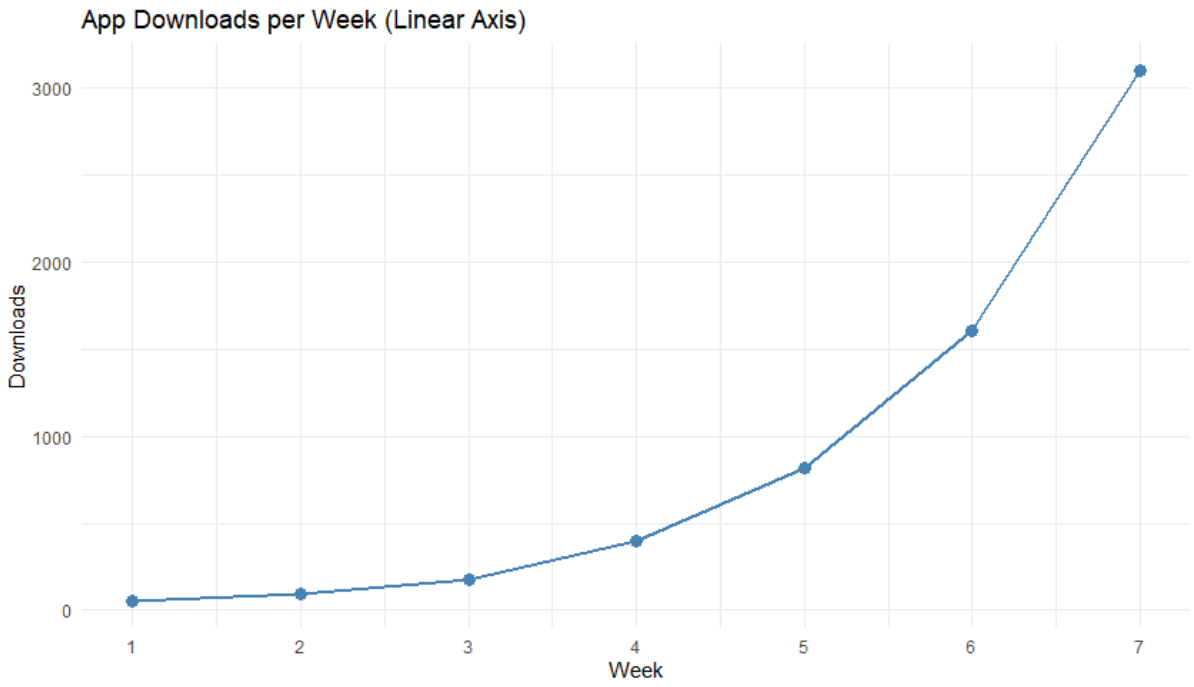
```
library(ggplot2)

downloads <- data.frame(
  Week = 1:7,
  Downloads = c(50, 95, 180, 400, 820, 1600, 3100)
)

ggplot(downloads, aes(x = Week, y = Downloads)) +
  geom_line(color = "steelblue", linewidth = 1) +
```

```
  geom_point(size = 3, color = "steelblue") +
  scale_x_continuous(breaks = 1:7) +
  labs(title = "App Downloads per Week (Linear Axis)",
       x = "Week", y = "Downloads") +
  theme_minimal()
```

Output :



2.2 Modified program with a logarithmic y-axis

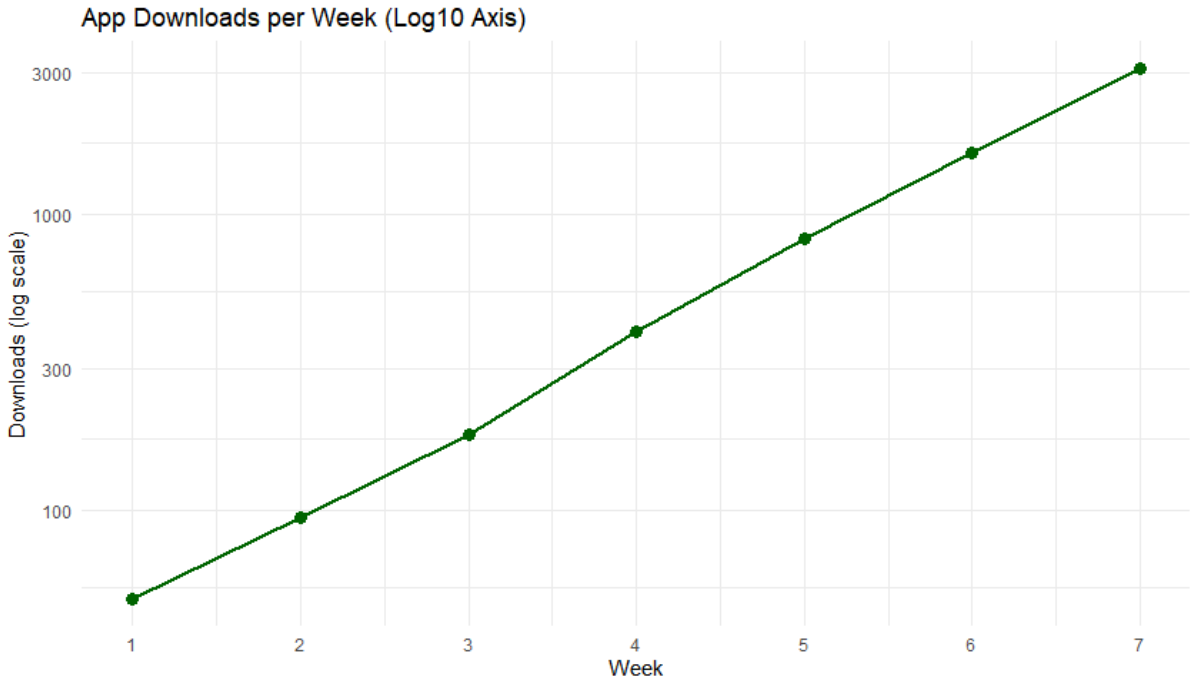
```
library(ggplot2)

downloads <- data.frame(
  Week = 1:7,
  Downloads = c(50, 95, 180, 400, 820, 1600, 3100)
)

ggplot(downloads, aes(x = Week, y = Downloads)) +
  geom_line(color = "darkgreen", linewidth = 1) +
  geom_point(size = 3, color = "darkgreen") +
  scale_x_continuous(breaks = 1:7) +
  scale_y_log10() +
  labs(title = "App Downloads per Week (Log10 Axis)",
```

```
    x = "Week", y = "Downloads (log scale)") +
theme_minimal()
```

Output:



2.3 Comparison of both visualizations

Aspect	**Linear y-axis**	**Logarithmic y-axis**
Shape of the curve	Curves sharply upward (exponential-looking)	Almost a straight line
Early weeks (1–3)	Points are squeezed near the bottom; differences are hard to see	Early points are clearly separated and readable
Later weeks (6–7)	Dominate the chart; large visual jumps	Shown in proportion with the early weeks
What the reader sees	Absolute increase in downloads	Relative (percentage) growth rate
Risk of misreading	Early growth appears negligible	Reader may think growth is slow unless they notice the log scale

In short, the linear plot shows how large the downloads become, while the log plot shows how fast they grow. The nearly straight line on the log plot shows that downloads roughly

double every week (95/50 ≈ 1.9, 180/95 ≈ 1.9, 400/180 ≈ 2.2, 820/400 ≈ 2.1, 1600/820 ≈ 1.95, 3100/1600 ≈ 1.9).

2.4 Why a logarithmic scale is suitable

The data grows exponentially and spans a wide range, from 50 to 3,100 (more than 60 times larger). On a linear axis the small values in weeks 1–3 are compressed and cannot be compared. A logarithmic scale gives equal vertical distance to equal ratios, so every doubling looks the same. This makes the pattern of steady proportional growth visible, shows the early weeks clearly, and makes it easy to compare growth rates between periods.

2.5 Situation where a logarithmic scale should not be used

A logarithmic scale should not be used when the data contains zero or negative values, because the logarithm of these values is undefined. Example: a company's weekly profit or loss, which can be negative. It is also unsuitable for a general audience that is not familiar with log axes, or when the data covers a narrow range (for example, exam marks between 60 and 90), since it adds complexity without benefit.

QUESTION 3

Question 3: Legends and Annotations [Python]

Dataset

Month	Company X (₹)	Company Y (₹)	Company Z (₹)
Jan	410	380	450
Feb	430	395	460
Mar	460	410	440
Apr	480	430	470
May	520	460	510
Jun	500	490	530

Questions

1. Write a Python program to plot the monthly stock-price trend for all three companies on a single chart with a clear legend.

2. Annotate (using plt.annotate()) the point of highest price for each company.

3. Explain the role of legends in multi-series visualizations.

4. Discuss best practices for placing legends to avoid obscuring data.

5. Suggest an alternative to a legend for distinguishing categories (e.g., direct labeling).

ANSWER 3

3.1 and 3.2 Python program (single chart, legend, and annotations)

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Company X": [410, 430, 460, 480, 520, 500],
    "Company Y": [380, 395, 410, 430, 460, 490],
    "Company Z": [450, 460, 440, 470, 510, 530]
```

```
}
df = pd.DataFrame(data)

colors = {"Company X": "tab:blue", "Company Y": "tab:orange",
          "Company Z": "tab:green"}

plt.figure(figsize=(9, 5))

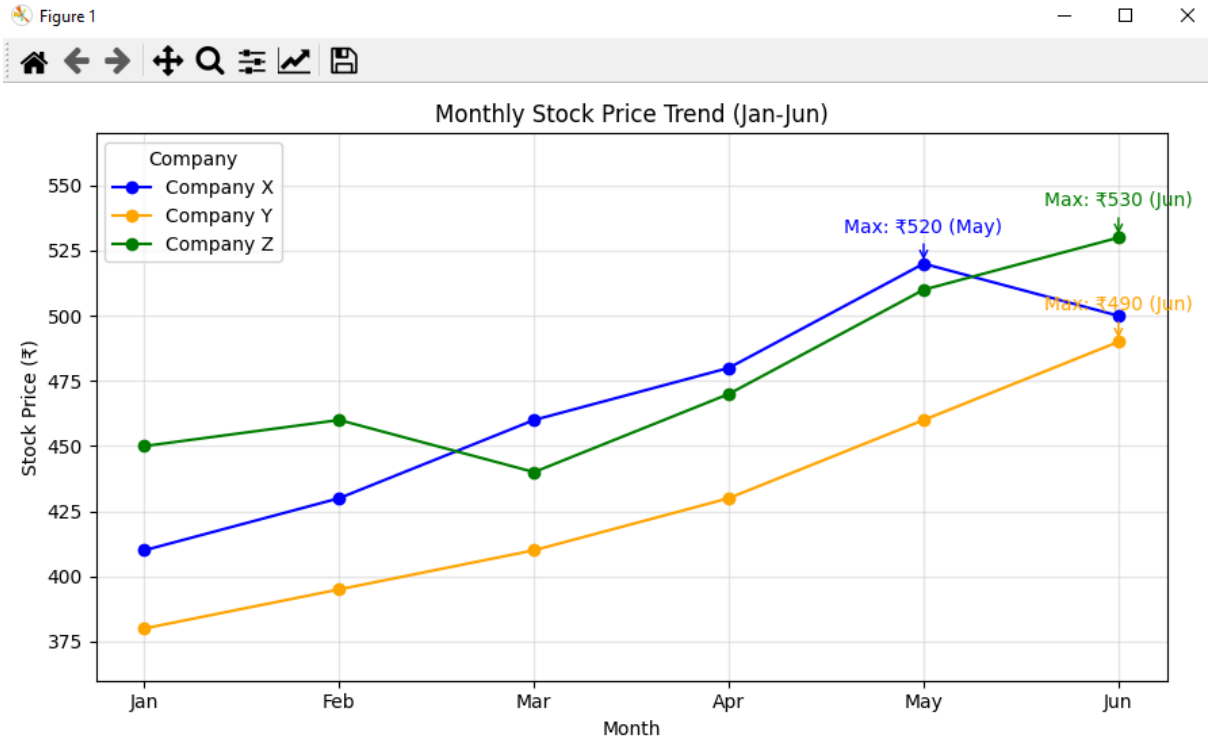
for company in ["Company X", "Company Y", "Company Z"]:
    plt.plot(df["Month"], df[company], marker="o",
             color=colors[company], label=company)

    # Find the highest price and annotate it
    idx = df[company].idxmax()
    month = df.loc[idx, "Month"]
    price = df.loc[idx, company]
    plt.annotate(f"Max: ₹{price} ({month})",
                 xy=(month, price),
                 xytext=(month, price + 12),
                 ha="center",
                 color=colors[company],
                 arrowprops=dict(arrowstyle="->", color=colors[company]))

plt.title("Monthly Stock Price Trend (Jan–Jun)")
plt.xlabel("Month")
plt.ylabel("Stock Price (₹)")
plt.ylim(360, 570)
plt.legend(title="Company", loc="upper left")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

Highest prices found by the program: Company X – ₹520 (May); Company Y – ₹490 (Jun); Company Z – ₹530 (Jun).

Output:



3.3 Role of legends in multi-series visualizations

A legend links each visual encoding (line color, marker, or style) to the series it represents. When several series share one chart, the reader cannot tell which line belongs to which company without it. A legend therefore makes the chart self-explanatory, allows quick comparison between series, and avoids confusion when colors or markers are similar. A legend with a title (for example “Company”) also states the category being compared.

3.4 Best practices for placing legends

- Place the legend in an empty area of the plot (for example the upper left, where prices are low in early months), using loc="upper left" or "best".
- Place it outside the plotting area (for example plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left")) when the chart is crowded.
- Keep the legend small: short labels, a small font, and a semi-transparent or no frame so that data points are not hidden.
- Order the legend entries in the same order as the lines, and use the same colors in the legend and the plot.
- Never place it over lines, markers, or annotations.

3.5 Alternative to a legend

Direct labeling is a good alternative: write each company's name at the end of its line (for example, plt.text("Jun", 500, "Company X") placed just to the right of the last point), in the same color as the line. The reader then does not need to look back and forth between the legend and the lines. Other alternatives are using small multiples (a separate panel for each company) or a table with colored markers placed next to the chart.

QUESTION 4

Question 4: Faceting / Small Multiples [R]

Dataset

Subject	Semester	Average Score
Maths	Sem1	72
Maths	Sem2	78
Maths	Sem3	81
Physics	Sem1	65
Physics	Sem2	70
Physics	Sem3	75
English	Sem1	80
English	Sem2	82
English	Sem3	85
CS	Sem1	74
CS	Sem2	83
CS	Sem3	90

Questions

1. Write an R program (using facet_wrap()) to create a faceted plot showing the average-score trend for each Subject separately.

2. Choose an appropriate variable to facet by and justify the choice.

3. Compare trends across the facets.

4. Explain the advantage of faceting over a single combined plot for this dataset.

5. Suggest a scenario where faceting would not be suitable.

ANSWER 4

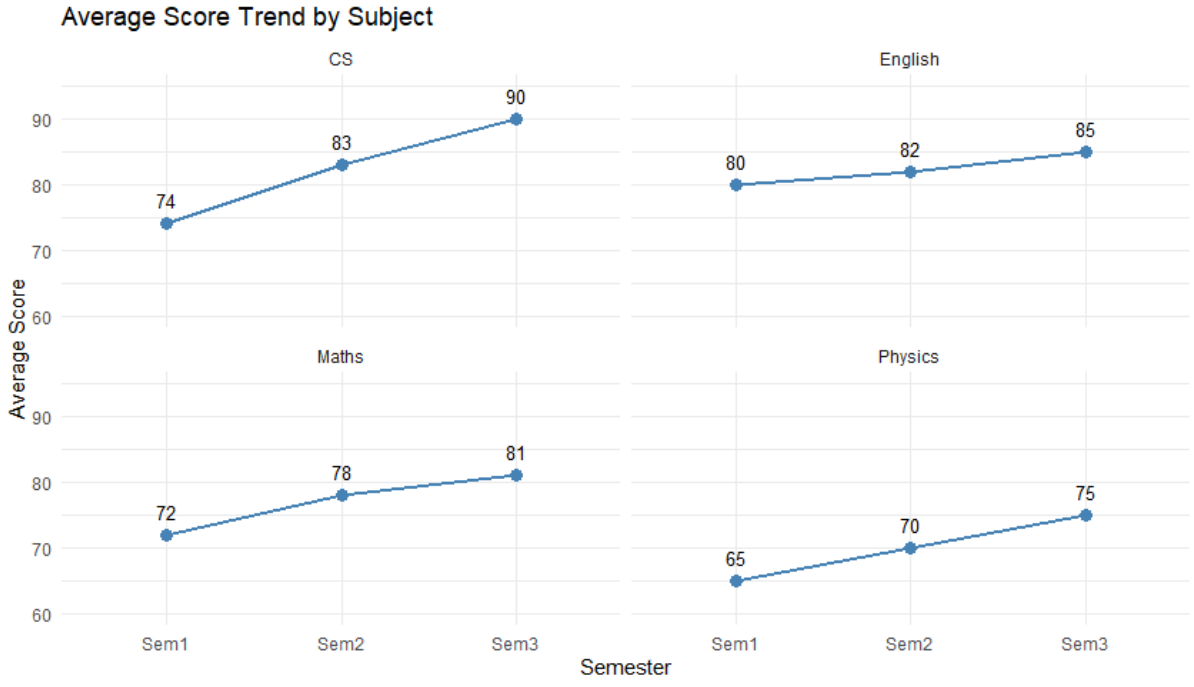
4.1 R program using facet_wrap()

```
library(ggplot2)
```

```
scores <- data.frame(
  Subject = rep(c("Maths", "Physics", "English", "CS"), each = 3),
  Semester = rep(c("Sem1", "Sem2", "Sem3"), times = 4),
  Score = c(72, 78, 81,
            65, 70, 75,
            80, 82, 85,
            74, 83, 90)
)

ggplot(scores, aes(x = Semester, y = Score, group = Subject)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(size = 3, color = "steelblue") +
  geom_text(aes(label = Score), vjust = -1, size = 3.5) +
  facet_wrap(~ Subject) +
  ylim(60, 95) +
  labs(title = "Average Score Trend by Subject",
       x = "Semester", y = "Average Score") +
  theme_minimal()
```

Output:



4.2 Variable chosen for faceting and justification

The variable chosen is Subject, written as facet_wrap(~ Subject). Justification:

- Subject is a categorical variable with only four levels, which gives a neat 2 × 2 grid of panels.
- The question asks for the trend of each subject separately, and the semester trend is the quantity to be compared inside each panel.
- Semester is the natural x-axis (ordered time), so it should not be used for faceting; faceting by Semester would hide the trend over time.
- All panels share the same y-axis limits, so scores remain directly comparable across subjects.

4.3 Comparison of trends across facets

Subject	**Sem1 → Sem3**	**Total change**	**Pattern**
Maths	72 → 81	+9	Steady increase (+6, then +3); growth slows
Physics	65 → 75	+10	Steady, even increase (+5, +5); lowest scores overall
English	80 → 85	+5	Small, gradual increase; highest in Sem1 and most stable
CS	74 → 90	+16	Fastest growth (+9, then +7); highest score in Sem3

All four subjects improve every semester. CS shows the largest improvement and ends with the highest score, Physics starts lowest but improves steadily, English is consistently high with the smallest change, and Maths improves but at a slowing rate.

4.4 Advantage of faceting over a single combined plot

In a single combined plot, four lines would be drawn on the same axes. Their values are close (65–90) and the lines would overlap and cross, so the reader would need a legend and effort to follow each line. Faceting gives each subject its own small panel with the same axes, so the shape of each trend is visible immediately, there is no overlap or clutter, no legend is needed, and subjects can still be compared easily because the scales are identical.

4.5 Scenario where faceting is not suitable

Faceting is not suitable when there are too many categories. For example, plotting the scores of 200 students would produce 200 tiny panels that are unreadable; a single plot with highlighted lines or a summary chart is better. It is also unsuitable when the main goal is a direct comparison of exact values between a few series on one axis (for example, to see where two lines cross), because the reader would need to compare across separate panels.

QUESTION 5

Question 5: Integrated Visualization [R]

Dataset

Country	GDP (₹ Trillion)	Literacy Rate (%)	Population (Million)	Continent
C1	12.5	92	140	Asia
C2	8.2	88	65	Europe
C3	21	95	330	N. America
C4	3.1	75	210	S. America
C5	5.6	80	95	Africa

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.
2. Select suitable aesthetics for GDP, Literacy Rate, Population, and Continent.
3. Interpret the relationship between GDP and Literacy Rate.
4. Identify the continuous and categorical variables.
5. Recommend one improvement to enhance the visualization's readability.

ANSWER 5

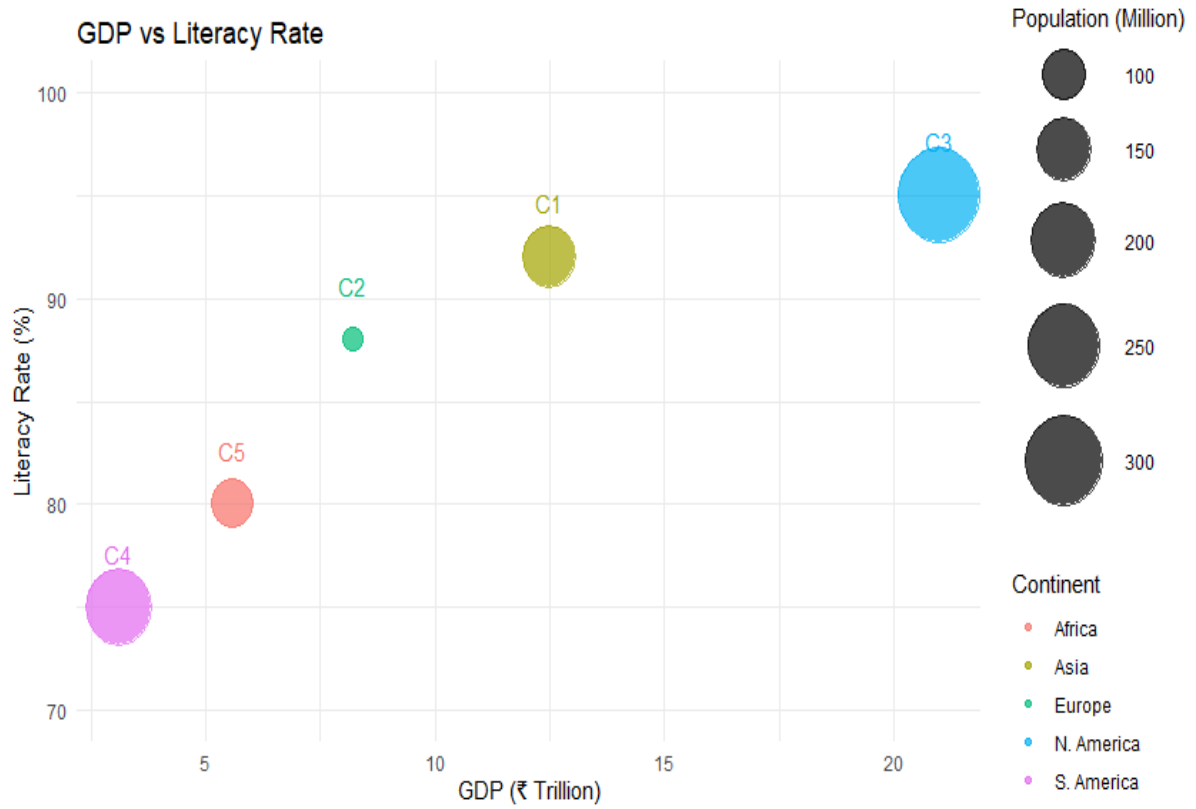
5.1 R program for the bubble chart

```
library(ggplot2)


countries <- data.frame(
  Country = c("C1", "C2", "C3", "C4", "C5"),
  GDP = c(12.5, 8.2, 21.0, 3.1, 5.6),
  Literacy = c(92, 88, 95, 75, 80),
  Population = c(140, 65, 330, 210, 95),
  Continent = c("Asia", "Europe", "N. America", "S. America", "Africa")
)
```

```
ggplot(countries, aes(x = GDP, y = Literacy,
                      size = Population, color = Continent)) +
  geom_point(alpha = 0.7) +
  geom_text(aes(label = Country), vjust = -2.2, size = 4,
            show.legend = FALSE) +
  scale_size(range = c(5, 20), name = "Population (Million)") +
  ylim(70, 100) +
  labs(title = "GDP vs Literacy Rate",
       x = "GDP (₹ Trillion)",
       y = "Literacy Rate (%)") +
  theme_minimal()
```

Output:



5.2 Suitable aesthetics

Variable	Aesthetic	Reason
GDP	x-axis position	Main numeric variable; position is the most accurate way to show a value
Literacy Rate	y-axis position	Second numeric variable; allows the relationship with GDP to be seen as a trend
Population	size (bubble area)	Third numeric variable; larger area means larger population
Continent	color (hue)	Categorical variable; distinct colors separate the groups

5.3 Relationship between GDP and Literacy Rate

There is a clear positive relationship: as GDP increases, the literacy rate increases. C4 (GDP 3.1, literacy 75%) and C5 (5.6, 80%) are at the lower left, C2 (8.2, 88%) and C1 (12.5, 92%) are in the middle, and C3 (21.0, 95%) is at the upper right. The increase is not perfectly linear: it is steep at low GDP values and flattens at high GDP values (for example, GDP rises from 12.5 to 21.0 but literacy rises by only 3 points). This shows diminishing returns. Population size does not follow the pattern (C4 has a large population but low literacy). Since there are only five countries, and correlation does not show causation, this conclusion is only indicative.

5.4 Continuous and categorical variables

Type	Variables
Continuous (numeric)	GDP (₹ Trillion), Literacy Rate (%), Population (Million)
Categorical	Continent (the Country column is only an identifier/label)

5.5 Recommended improvement for readability

Label each bubble directly with the country name (geom_text() or the ggrepel package's geom_text_repel()), so the reader does not have to match colors with a legend. Another improvement is to use transparency (alpha = 0.7), as already applied, so that overlapping bubbles remain visible. For larger datasets, scale_size_area() gives bubble areas that are proportional to population.